

Sakarya University, Department of Software Engineering
SWE 204 Concepts of Programming Languages, Spring 2026
Homework – 1

SABIS submission deadline: 1 April 2026, Wednesday, 21:00

Demo in class: 2 April 2026, Thursday, 09:00 – 12:00

For this assignment, you are expected to implement an LR parser for basic arithmetic expressions. Grammar, action and goto tables are provided in Figure 1. The LR parser you will develop must produce a trace table (similar to the ones we did in class) and a parse tree for each input expression.

State	Action						Goto		
	id	+	*	(	)	\$	E	T	F
0	S5			S4			1	2	3
1		S6				accept			
2		R2	S7		R2	R2			
3		R4	R4		R4	R4			
4	S5			S4			8	2	3
5		R6	R6		R6	R6			
6	S5			S4				9	3
7	S5			S4					10
8		S6			S11				
9		R1	S7		R1	R1			
10		R3	R3		R3	R3			
11		R5	R5		R5	R5			

1. $E \rightarrow E + T$
2. $E \rightarrow T$
3. $T \rightarrow T * F$
4. $T \rightarrow F$
5. $F \rightarrow (E)$
6. $F \rightarrow id$

Figure 1: Grammar, action table, and goto table for basic arithmetic expressions.

The following items attached in homework handout zip file:

1. **Grammar** file (Grammar.txt)
2. **Action** table (ActionTable.txt)
3. **Goto** table (GotoTable.txt)
4. **Test inputs** (input1.txt, input2.txt, ..., input9.txt): There are total 9 input files, each include an arithmetic expression.

Two out of total nine test inputs are not valid arithmetic expressions. More specifically, input7.txt includes syntax error and input9.txt includes an unknown token. Thus, the LR parser you will develop must behave accordingly to these erroneous inputs, i.e., print an error message and stop the parsing process.

The remaining seven test inputs are valid arithmetic expressions. The LR parser you will develop must accept and parse these test cases.

5. **Example outputs for each test case** (output1.txt, output2.txt, ..., output9.txt)

Each output file includes 1) a table for tracing parsing steps and stack states, and 2) a parse tree.

Your LR parser must produce an output file for each input as shown in these example output files.

6. **Example logs for selected test cases** (log1.txt, log3.txt, log7.txt, and log9.txt)

Total four log files are provided to demonstrate how a LR parser code flow works. The given log files correspond to input test cases 1, 3, 7, and 9.

Your code doesn't need to produce the exact log structure given to you, but producing logs that demonstrate the parsing process is suggested.

Notes:

- All tokens in Grammar, ActionTable, GotoTable, and input files are separated with a single space character.
- You can do this assignment individually or in a group of maximum three people.

Submission:

- Include a description of how to compile and test your LR parser.
- Include all group members' name and student ID in the submission.

In-class demo:

- Students will be asked to explain the logic of their code and demonstrate that their program works correctly. All students must be present for their homework demonstration.

Good luck.

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